

# What One Description Conveys to Each Observer.

An Observation Theory Companion to “Rate, Leakage, and Distortion for Gaussian Sources”

Andrew H. Bond  
San José State University

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## Abstract

A description of data is usually written for one reader and then held by others. Observation Theory treats every party that holds a description as an observer with its own view of the source, its own geometry for weighing errors, and its own budget, and it asks what the same description conveys to each of them. This supplement reads the results of a companion paper on Gaussian sources in that language. The paper’s theorems are stated and proved there in standard information theory, and nothing here adds to them. All of the results can be stated with standard source-coding objects. What this supplement adds is an interpretation. Observation Theory motivated four of the questions, and it reads their answers as statements about observers. What a description conveys to an observer is not fixed by what that observer sees. A tradeoff between the two costs requires the constraints to leave the encoder a choice among relevant error covariances. In the Gaussian model, the best one-dimensional description reads a direction that turns with the budget, which no single-party Gaussian problem with a fixed weighted quadratic distortion reproduces. And the description that conveys least to an observer at a given rate is the one that also describes what that observer sees. The rest of the paper, including its region and its convexity, is standard and owes nothing to the interpretation.

## 1 One description, two observers

An encoder sees a source and writes a description for a decoder that must reconstruct part of it. A second party holds the same description together with its own noisy view of the source, and it never decodes. Both parties are observers of one description. The decoder is the observer the description was written for. The second party is an observer the description was not written for, and what the description conveys to it is the information it gains beyond what it already sees.

Standard information theory has the pieces. The rate is what the decoder pays. What the second party gains is the conditional mutual information between the source and the description given its view, the quantity that secure source coding calls the leakage to an eavesdropper. The region of rates and leakages is known in single-letter form, and for Gaussian sources the companion paper [1] works out its structure.

Observation Theory asks a different question of the same objects. It asks how the conveyed content depends on who the observer is. The answer in this setting is that it depends on the observer’s geometry as well as on its data, and the companion paper’s theorems make that dependence exact. This supplement collects four results that Observation Theory motivated, states what each one says about observers, and marks plainly which parts of the paper owe nothing to this reading.

## 2 The decoder and the second party are both observers

In Observation Theory an observer is a triple  $O = (C, G, B)$ . The map  $C$  is what the observer reads of the source, the matrix  $G$  is the geometry in which it weighs errors, and  $B$  is its budget [2]. The content a description conveys to an observer that holds side information is the conditional mutual information between the source and the description given that side information, the second coordinate of the rate–content region of [2].

The companion paper has two observers. The decoder reads  $C_1 = F$ , the linear part  $Y = FT$  of the source it must reconstruct, weighs errors with  $G_1 = W$ , and has the distortion budget  $D$ . The second party, called the holder in the paper, reads  $C_2 = H$  through the noisy view  $S = HT + U$ , and the content conveyed to it is the leakage  $I(T; \hat{Y} \mid S)$ . The encoder sees the whole source  $T$ , which may include a context that neither observer’s reconstruction requires. The following table records the correspondence.

| Observation Theory                                    | Companion paper                              |
|-------------------------------------------------------|----------------------------------------------|
| observer the description is written for               | decoder, reading $Y = FT$ under weight $W$   |
| observer the description is not written for           | holder, seeing $S = HT + U$                  |
| content conveyed to an observer with side information | leakage $I(T; \hat{Y} \mid S)$               |
| budget of the first observer                          | distortion budget $D$                        |
| geometry of the second observer                       | information matrix $J = H^T \Sigma_U^{-1} H$ |
| what the encoder sees                                 | the source $T$ , context included            |

Every entry on the right is standard. Secure source coding introduced the leakage [4], and the vector Gaussian versions of the problem were treated by Ekrem and Ulukus [5] and by Xu, Chen, Chen, and Jin [6]. Renaming them does not create a result, and none of the four findings below depends on this vocabulary. Each can be stated, and is proved in the companion paper, with standard source-coding objects. The contribution of Observation Theory here is that it asked the questions, by treating the second party as an observer whose geometry matters, and that it gives the answers one coherent interpretation.

## 3 What a description conveys to an observer is not fixed by what the observer sees

One might expect the content conveyed to the holder to depend only on the joint law of what the decoder must reconstruct and what the holder sees. It does not. The scalar instances in Section X give two sources with the same joint law of  $(Y, S)$ , up to a scaling of  $S$ , whose least leakages at the same budget are 1.1880 and 1.2105 bits. The coupled instances give four such sources whose leakage-optimal descriptions reduce the error along  $42.83^\circ$ ,  $37.61^\circ$ ,  $33.75^\circ$ , and  $30.84^\circ$  at the same budget.

The difference is the encoder’s view. The encoder sees a context that neither observer reconstructs, and the context is correlated with what the holder sees. So the content conveyed to the second observer is a function of three things, the first observer’s task, the second observer’s view, and what the writer of the description sees. The joint law of the two observers’ views alone does not determine it. This is the reason the companion paper’s setting has an encoder that observes more than it describes.

## 4 A tradeoff requires the constraints to leave a choice among error covariances

Suppose the decoder's requirement is a full error covariance on the whole source, so that every direction must be reconstructed to a prescribed accuracy. The content conveyed to the holder is still a cost, and it still depends on the holder. But Proposition 4 shows that one description minimizes the rate and that cost together, so the two costs do not conflict. The result is implied by the bounds of Ekrem and Ulukus [5].

The two costs conflict when the constraints leave the encoder a choice among error covariances that matter to the holder. A weighted total distortion leaves such a choice, because the encoder decides how to spread the error over directions. So does a matrix constraint on the described part alone, which leaves the error free outside that part, as the companion paper notes after Proposition 4. In the language of Observation Theory, the first observer's constraints must leave freedom that the second observer's geometry can act on. Every other result below lives in that freedom.

## 5 The observer's geometry changes what an optimal description reads

This is the central finding. Its scope is the Gaussian model and the best one-dimensional description, which is the leakage-optimal description wherever the certificate of the companion paper holds.

A description written for a single observer under a weighted quadratic distortion reads fixed directions of the source at every budget. The budget sets how many directions are described and how finely, never which ones (Proposition 19). Principal components, reverse water-filling, and the Gaussian information bottleneck all behave this way.

When a second observer is present, the best one-dimensional description reads a direction that turns as the budget changes, unless that direction is shared by the decoder's geometry and the holder's (Theorem 20, which assumes the largest eigenvalue of its pencil is simple). In the coupled instance the leakage-optimal description reduces the error along  $47.48^\circ$  at one budget and  $41.15^\circ$  at another. Corollary 23 states what this excludes. No single-party Gaussian rate-distortion problem with a fixed weighted quadratic distortion has the leakage-optimal descriptions as its optimal descriptions at two such budgets. It does not exclude every richer single-observer model, such as one whose distortion measure changes with the budget. Within the stated class, the turning is the trace of the second observer.

The cost of ignoring it is explicit. A description that keeps a fixed direction pays an excess in content conveyed to the holder that Proposition 24 and Theorem 25 bound from below. For a scalar source with a nearly clean context, every fixed direction conveys at least 0.0245 bits more than necessary at one of two budgets, which exceeds the least leakage 0.0216 bits at the second budget.

The turning comes from the misalignment of the two observers' geometries, not from the context. A described variable with no context at all turns whenever the decoder's weight and the holder's information share no eigenvector.

## 6 To convey little to an observer, describe what that observer sees

The general solution has a reading that is natural in Observation Theory. Theorem 7 shows that every frontier description is reverse water-filling on the decoder's weight tilted by the information

the holder’s view adds. Corollary 8 restates this. The description that conveys least to the holder at a given rate is the rate–distortion optimal description for a decoder that must reconstruct the holder’s own view as well as its target, with a weight that the solution itself fixes.

The reason is an identity. The content conveyed to the holder is the rate minus the information the description carries about what the holder sees. So a description that already tells the holder much of what it sees has little left to convey beyond it. In the language of the theory, the first observer’s effective geometry is its own geometry plus a copy of the second observer’s view, weighted by how uncertain that view remains after the description.

For a scalar described variable this becomes a single explicit curve (Theorem 18). Every optimal description, at every budget and every weighting of the two costs, reads a direction on the path from the decoder’s own target toward the target corrected by the holder’s information. The rate-optimal description sits at one end, and the description that conveys least sits at a point that moves with the budget.

## 7 On eleven datasets the single-party flip holds, and the two-observer flip holds where the Gaussian model does

The companion notebook `ot_figures/ot_figures.ipynb` tests two flips on eleven public datasets, including the Topology Zoo table of a routing study. For each dataset the roles were fixed before any outcome was computed and pushed to the public repository on 2 October 2026, and the run started a minute after the push, as the notebook records. Predictions use the second-order statistics of a training half. Outcomes use real 3-bit quantizers on a held-out half. On real data the conditional leakage  $H(M | S)$  of a code  $M$  cannot be computed, and every measured leakage in this section is a proxy for it, the cross-entropy of the best of three classifiers  $q(m | s)$  that predict the code from the second party’s view. Its population value is

$$\mathbb{E}[-\log_2 q(M | S)] = H(M | S) + \mathbb{E}_S D(p(M | S) \| q(M | S)),$$

so it adds the classifier’s error to the leakage, and the difference of two proxies mixes a difference of conditional entropies with a difference of classifier errors. In these runs the classifier was also chosen on the same held-out samples on which its loss was reported. The ordering of two proxies therefore does not establish the ordering of the true conditional entropies, and the claims below are claims about the proxy.

The single-party flip holds in all eleven datasets. At matched bits, the code that reconstructs the whole source worse serves the decoder better. Its boundary holds as predicted. Both halves of the flip vanish as the angle between the decoder’s task and the source’s energy direction goes to zero, and grow with it (Fig. 1), although the whole-source half is noisy below 30 degrees.

The two-observer flip holds at all three budgets in eight datasets and at two of three in a ninth. At matched bits, the code that reconstructs the decoder’s target worse has the lower leakage proxy at the second party. It fails in two datasets, a wine-quality table with a discrete target and the 175-row Topology Zoo table, where the leakage-aware code had the higher proxy. Across 115 choices of what the second party sees, the predicted correlation ranks the leakage removed with a Spearman correlation of 0.62, with a residual at zero correlation. Two further predictions were refuted and are recorded in the notebook. A view uncorrelated with the decoder’s principal read still yields a flip once the budget has slack, and a local onset test held in only four of eleven datasets. Both used the squared correlation with the decoder’s principal read as the predictor. The boundary theory below replaces it with the position of that read in a pencil of the view, under which an uncorrelated view can still flip once the budget has slack.

The boundary of the two-observer flip was then registered in two stages. The theory places it at the position of the decoder’s principal read in a pencil of the second party’s view. If that read is the view’s most predictable direction there is no flip at any slack. Otherwise the flip appears, with a size the Gaussian model predicts. Predictions for 561 cases were pushed with the scoring script before any held-out code was measured. They cover thirteen sources (the eleven datasets and two synthetic Gaussian ones), each context column as the second party’s view, a noisy copy of the principal read as a control, and a family of views rotated away from that read. Two of the four registered criteria passed. Where the predicted removal is at least 0.10 bit, the measured removal is positive in 81 percent of 413 cases, and the Spearman correlation over all cases is 0.67. On the synthetic Gaussian sources every case agrees with its prediction within two standard errors, and the control measures zero. Two criteria were not met. Where the predicted removal is below 0.05 bit, 56 percent of 57 cases fall within two standard errors of zero, against a registered 80 percent. Within the rotated family the rank correlation is 0.56, against a registered 0.6. Nine of the eleven datasets track the prediction with rank correlations between 0.81 and 0.98. The bike-rental table, whose targets are counts, ranks its cases correctly at a tenth of the predicted size. The wine and Topology Zoo tables are again reversed. We read these misses as returns from the edge of the Gaussian model, and a second registration tests that reading by asking whether a measured departure from Gaussianity predicts where they fall.

The leakage predictions rest on one modelling step. What the second party learns from a code depends on its view only through a linear correlation, which is exact when the projection and the view are jointly Gaussian. The second registration measures, on training halves only, how far each code departs from that step. It adds the information the view carries beyond the linear correlation to the divergence of the code’s cell occupancies from their Gaussian design, each in bits above the 95th percentile of a Gaussian null of the same size. On the measured datasets this index ranks the size of each miss with a Spearman correlation of 0.71 over 477 cases. It was then applied blind to six public datasets this program had never touched, with a cut calibrated on the eleven measured ones. Before any of their codes were measured, it forecast that a power-plant table, an airfoil-noise table, and a real-estate table would track the theory, and that a yacht-hydrodynamics table, a forest-fire table, and a red-wine table would not. All six forecasts were right. On the fresh cases the index ranks the misses with a correlation of 0.67, and the cases it cannot tell from Gaussian agree with their predictions in 16 of 16, against 33 percent of the rest. Two criteria were not met. A calibration bar asked that 90 percent of the synthetic Gaussian cases score zero, and 84 percent did. Each case faces four tests at the 95th percentile, so the bar should have been near 81 percent. A second comparison was vacuous, because no case from the measured datasets scored zero. The theory’s own modelling assumption thus forecasts where the theory holds. Read after the fact and not as a test, the two criteria the boundary test missed behave the same way. Below the cut the rank correlation in the rotated family is 0.77 on the measured datasets and 0.74 on the fresh ones. Above it the correlation is 0.31 and  $-0.15$ . The near-boundary criterion is still unmet below the cut on the measured datasets, at 64 percent.

A third registration tested these readings blind on six more public datasets, with the departure cut frozen in advance. The forecast was again right for all six. Inside the Gaussian scope the measured removal was positive in all 180 cases predicted at 0.10 bit or more, the rank correlation was 0.97, and the rotated family reached 0.92, so the criterion the boundary test missed holds inside the scope on new data. The onset criteria passed, the second-order family staying at zero at the smallest slack in 4 of 4 cases and the first-order family rising above it in 4 of 5. The near-boundary criterion failed for the third time, at 35 percent of 23 cases, so it fails inside the scope as well as outside it. The registration also tested the deduced account of the misses. The error of a leakage prediction, measured against the true conditional entropy, splits exactly into a marginal defect, the

mismatch between a code’s cell occupancy and its Gaussian design, and a dependence defect, the information the view carries beyond its linear correlation. The proxy adds the classifier’s error to both. On the 20 earlier sources the marginal defect accounted for most of the error. On the new data, correcting for it did not lower the mean miss, which was 0.096 bit against 0.092 without the correction. Codes with equal cell occupancy restored the ranking on the one dataset outside the scope, from  $-0.34$  to  $0.82$ , but agreed with their predictions less often than the standard codes outside the scope, in 9 percent of cases against 23. A symbolic-regression search over the earlier sources had found the squared difference of the two codes’ dependence defects as the predictor of the remaining misses. It did not hold across datasets there and was only reported, but on the new data it classified 86 percent of cases correctly, against 63 percent for always predicting no miss. Outside the Gaussian scope the dependence defect, or the classifier error that the proxy adds, is therefore the larger unexplained term, and the marginal account of the misses holds only in part.

A fourth registration asked why the near-boundary criterion keeps failing. One account held that its standard error, a bootstrap over the held-out points of a single split, understates the variability of the measurement. The other held that the measured removal is systematically nonzero near the boundary. The 71 in-scope near-boundary cases on real data were measured again on 20 fresh splits. The spread across splits was smaller than the bootstrap error, at a median ratio of 0.76, so the first account fails. In half the cases the mean over the splits sat more than three standard errors from zero, so by the reading fixed in advance the misses are systematic. A comparison made after the scoring shows what the criterion had been testing. Its cases are not predicted at zero. Their predicted removals lie between 0.03 and 0.05 bit, and the criterion compared the measurement with zero rather than with the prediction. On synthetic Gaussian sources the mean over splits lies within 0.006 bit of the prediction, and every single measurement is within two standard errors of it. The replicate splits of this registration had a design error, found later. They recomputed the principal read and the context column of the rotated views on every split, although both were meant to be frozen, and on most real datasets that column changes between splits. A rotated-view case was then measured against a view its code was not built for. On the 20 real cases whose view is a single context column, which the error does not touch, the mean over splits falls short of the prediction by a median of 0.010 bit, in 90 percent of them by more than three standard errors. The larger shortfall of about 0.05 bit on the 51 rotated-view cases comes mostly from the error. So on Gaussian data the proxy matches the prediction near the boundary, and on real data inside the scope it falls slightly but systematically short of it. These runs do not separate a misspecified Gaussian model from unequal classifier errors on the two codes, and the cause of the shortfall is open. A fifth registration tests the corrected criterion against the prediction, and whether the shortfall on real data is a fixed offset or a proportional attenuation. Its replicate splits carried the same error, and both replicate designs are being measured again with the directions frozen as registered.

**Experiment ledger.** The companion notebook covers the first campaign only. Table 1 lists every registration with the commit at which its code and predictions were frozen and the commit of its scored outcome, all in the public repository under `paper/tit-rate-leakage`.

The deduction of the two defects (`DEDUCTION-gaussian-defect-2026-10-02.md`, `c9fcc18`) was committed before it was computed. The mining and symbolic-regression runs that followed it (`ot_mining.py`, `ot_radar_induction.py`) are exploratory and are not registrations.

Flip A against the predicted angle, eleven datasets

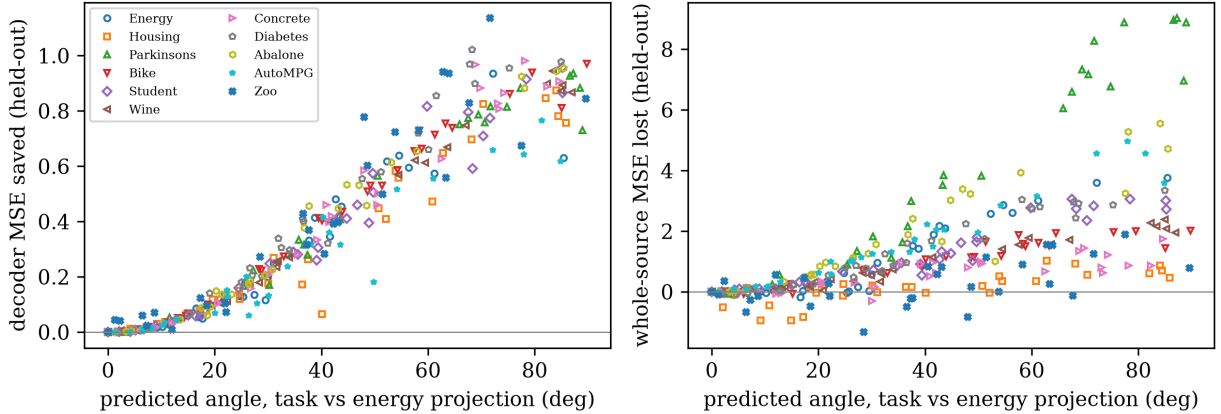


Figure 1: The single-party flip on eleven datasets. Each point is one decoder task. The decoder’s gain (left) and the whole-source loss (right) of the consumer code are both near zero where the training half predicts alignment, and grow with the predicted angle.

## 8 What does not need Observation Theory

Most of the companion paper is standard information theory, and its value there does not depend on the reading above. The single-letter region is due to Villard and Piantanida [4]. Without decoder side information it is a union of the matrix-distortion regions of Ekrem and Ulukus [5], and the paper proves the Gaussian coding theorem directly (Theorem 3). The convexity of both costs in the error covariance (Theorem 5), the dimension bound (Proposition 6), the closed forms in commuting geometry and at low distortion (Theorem 13, Theorem 15), the one-dimensional pencil (Theorem 16), and the monotone iteration (Proposition 11) are results about a convex program. They would be stated the same way by a reader who had never heard of observers.

## 9 Open questions about several observers

The companion paper has one observer the description is written for and one it is not. Observation Theory asks the same questions of more.

*Several holders.* With holders that see different views, the content conveyed to each is a separate cost, and the frontier becomes a surface. A weighted Gaussian objective already gives a sum of tilts, one per holder, by the argument of Theorem 7. The open questions are operational and geometric. Whether every point of that surface is achievable needs a coding theorem for several holders at once, and whether it has explicit solutions, for instance a direction that turns toward one holder and away from another, is not known.

*Observers on both sides.* When the decoder also has side information and neither side dominates, the second auxiliary of the secure source coding region can matter. Xu, Chen, Chen, and Jin [6] characterize the region under a matrix distortion on the whole source, stated in the preprint for full-rank views of both receivers. Whether the observer effects of Sections 5 and 6 survive under constraints that leave a choice of error covariance, such as a weighted total distortion, and a rank-deficient holder is the main open question.

*Observers with rich views.* For a holder whose view has rank above one and a described variable

Table 1: Registrations of the empirical campaign. Verdicts are as scored by the frozen scripts. Every leakage is the classifier cross-entropy proxy described in Section 7.

| Registration                 | Frozen code and predictions                                                      | Scoring and outcome                  | Verdicts                                                                                              |
|------------------------------|----------------------------------------------------------------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------|
| Flips A and B                | ot_figures.ipynb, roles on branch prespec/ot-notebook-2026-10-01                 | notebook record, 9ffbd20             | Flip A 11 of 11, Flip B 8 of 11, onset refuted                                                        |
| Boundary of Flip B           | ot_boundary.py, boundary_predictions.json, 7332b8c                               | boundary_score.py (86eccd7), daca788 | C1 pass, C2 fail, C3 pass, C5 fail                                                                    |
| Gaussian departure           | ot_gaussianity.py (1441065), gauss_predictions.json (ec3300d)                    | gauss_score.py, 9f2a61b              | R1 pass, R2 vacuous, K1 fail, B1 pass, B2 pass, B3 6 of 6                                             |
| Gaussian scope               | ot_scope.py (74c446d), scope_predictions.json (9f5f2ca), correction note 15cca7b | scope_score.py, c14e579              | S0 6 of 6, S1 pass, S2 fail, S3 pass, S4a pass, S4b pass, K1 pass, M1 fail, E1 fail, E2 fail, E3 pass |
| Near-boundary noise          | ot_s2noise.py (fec42f0)                                                          | built into the script, 9fa0d23       | N1 fail, N2 fail, B1 pass; rotated views confounded by a design error, re-run registered              |
| Corrected near-boundary test | ot_nearboundary.py (ae487c4), nb_predictions.json (3f9d1aa)                      | built into the script, d5ff89c       | P1c pass, P1r fail, SH pass, D1 fail, A1 pass; confounded by the same design error, re-run registered |
| Design correction            | ot_crossfit_c.py                                                                 | built into the script                | pending                                                                                               |

of dimension above one, the frontier is the limit of an iteration of closed-form steps (Proposition 11) but has no known closed form.

*Beyond Gaussian sources.* The non-determination of Section 3 and the turning of Section 5 were found in the Gaussian case. Whether they hold for finite alphabets, where the single-letter region is also known [4], is open.

## Where this fits

The observer triple and the rate-content region are defined in [2], which also treats two observers on one description. The vector form of the two-observer results, with several described coordinates, is in [3]. The scripts that check every number quoted here are listed in the companion paper’s appendix on numerical checks.

## References

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